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Results in Mathematics



Some Congruences from the Karlsson-Minton Summation Formula

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Abstract. Let p be an odd prime. In this paper, by using the well-known Karlsson-Minton summation formula, we mainly prove two supercongruences as variants of a supercongruence of Deines-Fuselier-Long-Swisher-Tu, which confirm some recent conjectures of V.J.W. Guo.

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1. Introduction

For any $n \in \mathbb{N} = \{0, 1, 2, \ldots\}$, let $(x)_n = x(x+1)\cdots(x+n-1)$ denote the Pochhammer symbol. For $n, r \in \mathbb{N}$ and $a_0, \ldots, a_r, b_1, \ldots, b_r, z \in \mathbb{C}$ with $(b_1)_n, \ldots, (b_r)_n$ being nonzero, the truncated hypergeometric series ${}_{r+1}F_r$ are defined as

$$ {}_{r+1}F_r\left[\begin{matrix} a_0, a_1, \ldots, a_r \\ b_1, \ldots, b_r \end{matrix}\middle| z\right]_n = \sum_{k=0}^{n} \frac{(a_0)_k \cdots (a_r)_k}{(b_1)_k \cdots (b_r)_k} \cdot \frac{z^k}{k!}. $$

Clearly, they are partial sums of the classical hypergeometric series. Let p be an odd prime. For any integer $n \geq 1$, the p-adic Gamma function introduced by Morita (cf. [8,12]) is defined as

$$ \Gamma_p(n) = (-1)^n \prod_{\substack{1 \leq k < n \\ p \nmid k}} k. $$

Moreover, set $\Gamma_p(0)=1$, and for any p-adic integer x set

$$\Gamma_p(x)=\lim_{n\to x}\Gamma_p(n),$$

where n runs through any sequence of positive integers p-adically approaching x.

Rodriguez-Villegas [13] investigated hypergeometric families of Calabi-Yau manifolds, and discovered (numerically) a number of possible supercongruences. Some of them have been proved in [9,10] where Mortenson, with the help of the Gross-Koblitz formula, determined ${}_2F_1\left[\begin{matrix}\alpha, 1-\alpha\\ 1\end{matrix}\middle| 1\right]_{p-1}$ modulo p^2 for $\alpha\in\{1/2,1/3,1/4,1/6\}$. For instance, he showed that

$$ {}_2F_1\left[\begin{matrix}\frac{1}{2}, \frac{1}{2}\\ 1\end{matrix}\middle| 1\right]_{p-1}\equiv(-1)^{(p-1)/2}\pmod{p^2} \tag{1.1}$$

for any prime $p\geq 5$. Using the Legendre relation of p-adic Gamma function (cf. [12, p. 370]), we may replace the right-hand side of (1.1) with $-\Gamma_p(1/2)^2$. Later, Sun [14] extended Mortenson's result to the general p-adic integer α. Let $\mathbb{Z}_p$ denote the ring of all p-adic integers and $\mathbb{Z}_p^\times:=\{x\in\mathbb{Z}_p: p\nmid x\}$. Z.-H. Sun proved that for each odd prime p and $\alpha\in\mathbb{Z}_p^\times$,

$$ {}_2F_1\left[\begin{matrix}\alpha, 1-\alpha\\ 1\end{matrix}\middle| 1\right]_{p-1}\equiv(-1)^{\langle-\alpha\rangle_p}\pmod{p^2}, \tag{1.2}$$

where $\langle x\rangle_p$ is the least nonnegative residue of x modulo p, i.e., $0\leq\langle x\rangle_p\leq p-1$ and $x\equiv\langle x\rangle_p\pmod p$. On the other hand, Deines et al. [1] obtained the following generalization of (1.1): for any integer $d>1$ and prime $p\equiv 1\pmod d$,

$$ {}_dF_{d-1}\left[\begin{matrix}1-\frac{1}{d}, 1-\frac{1}{d}, \ldots, 1-\frac{1}{d}\\ 1, \quad \ldots, \quad 1\end{matrix}\middle| 1\right]_{p-1}\equiv-\Gamma_p\left(\frac{1}{d}\right)^d\pmod{p^2}. \tag{1.3}$$

In fact, Deines et al. also conjectured that for any integer $d\geq 3$ and prime $p\equiv 1\pmod d$, the congruence (1.3) holds modulo p^3, and this conjecture was later confirmed by the third author and Pan [15].

In the past decade, many mathematicians studied q-analogues of (1.1) and its generalizations; among these q-congruences, the first one was obtained by Guo and Zeng [4]. Recently, via the so-called 'creative microscoping' method introduced by Guo and Zudilin [5], Guo [3] established a q-analogue of (1.3). Meanwhile, Guo obtained a variant of (1.3) as follows: for any integer $d>1$ and prime $p\equiv 1\pmod d$,

$$\sum_{k=0}^{p-1}\frac{k\left(\frac{d-1}{d}\right)_k^d}{k!^d}\equiv\frac{(d-1)\Gamma_p\left(\frac{1}{d}\right)^d}{2d}\pmod{p^2}.$$

The main purpose of this paper is to prove the following variants of (1.3) which confirm two conjectures of Guo [3, (5.4) and (5.5)].

Theorem 1.1. (i) *Let $d \geq 4$ be an even integer. Then, for any prime $p \equiv -1 \pmod{d}$ with $p \geq 2d-1$,*

$$ {}_dF_{d-1}\left[\begin{matrix}\frac{1}{d}-1, 1+\frac{1}{d}, 1+\frac{1}{d}, \ldots, 1+\frac{1}{d} \\ 1, \quad 1, \quad \ldots, \quad 1\end{matrix}\middle| \, 1\right]_{p-1} \equiv \frac{d-1}{d^2}\Gamma_p\left(-\frac{1}{d}\right)^d \pmod{p^2}. \tag{1.4}$$

(ii) *Let $d \geq 3$ be an odd integer. Then, for any prime $p \equiv -1 \pmod{d}$,*

$$ {}_dF_{d-1}\left[\begin{matrix}\frac{1}{d}, \frac{1}{d}, 1+\frac{1}{d}, 1+\frac{1}{d}, \ldots, 1+\frac{1}{d} \\ 1, \quad 1, \quad 1, \quad \ldots, \quad 1\end{matrix}\middle| \, 1\right]_{p-1} \equiv -\frac{1}{d^2}\Gamma_p\left(-\frac{1}{d}\right)^d \pmod{p^2}. \tag{1.5}$$

The second goal is to show another conjectural congruence [3, Conjecture 1.3].

Theorem 1.2. *Let $p \equiv 1 \pmod{4}$ be a prime and $r \geq 1$. Then*

$$\sum_{k=0}^{p^r-1}\left(k-\frac{p^{2r}-1}{4}\right)\frac{\left(\frac{1}{2}\right)_k^2}{k!^2} \equiv 0 \pmod{p^{2r+1}}. \tag{1.6}$$

The rest of this paper is organized as follows. In the next section, we list some necessary lemmas which play key roles in the proof of Theorem 1.1. Section 3 is devoted to the proof of Theorem 1.1. In Sect. 4, we prove Theorem 1.2. In Sect. 5, we shall pose a conjecture for further research.

2. Some necessary lemmas

The first key ingredient of our proofs is the following Karlsson-Minton summation formula (cf. [2, p. 19]).

Lemma 2.1. *Let $m_1, m_2, \ldots, m_n$ be nonnegative integers. Then*

$$\begin{aligned} &{}_{n+1}F_n\left[\begin{matrix}-(m_1+\cdots+m_n), b_1+m_1, \ldots, b_n+m_n, \\ b_1, \quad \ldots, \quad b_n\end{matrix}\middle| \, 1\right] \\ &\quad = (-1)^{m_1+\cdots+m_n}\cdot\frac{(m_1+\cdots+m_n)!}{(b_1)_{m_1}\cdots(b_n)_{m_n}}. \end{aligned} \tag{2.1}$$

Our proofs also rely on some properties of the p-adic Gamma functions.

Lemma 2.2. [12, p. 369] *Let p be an odd prime and $x \in \mathbb{Z}_p$. Then*

$$\frac{\Gamma_p(x+1)}{\Gamma_p(x)} = \begin{cases} -x, & \text{if } p \nmid x, \\ -1, & \text{if } p \mid x, \end{cases} \tag{2.2}$$

$$\Gamma_p(x)\Gamma_p(1-x) = (-1)^{\langle -x\rangle_p - 1}. \tag{2.3}$$

Remark 2.1. (a) By (2.2), it is easy to see that for any positive integer $n \leq p$,

$$\Gamma_p(n) = (-1)^n\Gamma(n). \tag{2.4}$$

(b) The identity (2.3) is a p-adic analogue of the following Legendre relation of the classical Gamma function:

$$\Gamma(x)\Gamma(1-x)=\frac{\pi}{\sin\pi x}.$$

The next lemma concerns a p-adic approximation to Γ_p-quotients.

Lemma 2.3 [7, Theorem 14]. *For any prime $p\geq 5$ and $x\in\mathbb{Z}_p$, there exists $G_1(x)\in\mathbb{Z}_p$ such that for any $t\in\mathbb{Z}_p$,*

$$\Gamma_p(x+tp)\equiv\Gamma_p(x)(1+G_1(x)tp)\pmod{p^2}. \tag{2.5}$$

For the properties of $G_1(x)$, the reader may consult [11]. The following lemma lists two identities involving the derivatives of $(1+\alpha+x)_k$, which can be verified directly.

Lemma 2.4. *For any integer $k\geq 0$ and $\alpha,\beta\in\mathbb{R}$,*

$$\frac{d}{dx}(1+\alpha+x)_k=(1+\alpha+x)_k\sum_{j=1}^{k}\frac{1}{j+\alpha+x},$$

$$\frac{d}{dx}\left(\frac{1}{(1+\beta+x)_k}\right)=-\frac{1}{(1+\beta+x)_k}\sum_{j=1}^{k}\frac{1}{j+\beta+x}.$$

3. Proof of Theorem 1.1

Throughout this section, we set $m=(p+1)/d$. We first prove (1.4). To show (1.4), we need the following preliminary result.

Lemma 3.1. *Under the assumptions of Theorem 1.1 (i), modulo p, we have*

$$\sum_{k=0}^{p-1}\frac{(m-1)_k(m+1)_k^{d-1}}{(1)_k^d}\left(\sum_{j=0}^{k-1}\frac{1}{m-1+j}-\sum_{j=0}^{k-1}\frac{1}{m+1+j}\right)$$
$$\equiv\frac{(p-1)!}{(1)_{m-2}(1)_m^{d-1}}\left(\frac{1}{m}+\frac{1}{m-1}\right).$$

Proof. For $x,y\in(-1,+\infty)$, set

$$\Psi(x,y)={}_{d+1}F_d\left[\begin{matrix}1-p, & m-1+x, & m+1+y, & m+1, & \ldots, & m+1\\ & 1+x, & 1+y, & 1, & \ldots, & 1\end{matrix}\,\middle|\,1\right]_{p-1}. \tag{3.1}$$

Clearly, $\Psi(x,y)$ is smooth on $(-1,+\infty)\times(-1,+\infty)$. Since $(1-p)_k=0$ for all $k\geq p$, we have

$$\Psi(x,y)={}_{d+1}F_d\left[\begin{matrix}1-p, & m-1+x, & m+1+y, & m+1, & \ldots, & m+1\\ & 1+x, & 1+y, & 1, & \ldots, & 1\end{matrix}\,\middle|\,1\right].$$

As

$$m-2+(d-1)m=m-2+p+1-m=p-1,$$

by Lemma 2.1,

$$\Psi(x,y)=\frac{(p-1)!}{(1+x)_{m-2}(1+y)_m(1)_m^{d-2}}. \tag{3.2}$$

Now we calculate $\Psi_x(0,0)-\Psi_y(0,0)$ in two different ways, where $\Psi_x(0,0)$ and $\Psi_y(0,0)$ stand for the partial derivatives of Ψ at $(0,0)$ with respect to x and y. By (3.1) and Lemma 2.4, we obtain

$$\Psi_x(0,0)=\sum_{k=0}^{p-1}\frac{(1-p)_k(m-1)_k(m+1)_k^{d-1}}{(1)_k^{d+1}}\left(\sum_{j=0}^{k-1}\frac{1}{m-1+j}-H_k\right),$$

$$\Psi_y(0,0)=\sum_{k=0}^{p-1}\frac{(1-p)_k(m-1)_k(m+1)_k^{d-1}}{(1)_k^{d+1}}\left(\sum_{j=0}^{k-1}\frac{1}{m+1+j}-H_k\right),$$

where $H_k=\sum_{j=1}^{k}1/j$ denotes the harmonic number. Therefore,

$$\Psi_x(0,0)-\Psi_y(0,0)=\sum_{k=0}^{p-1}\frac{(1-p)_k(m-1)_k(m+1)_k^{d-1}}{(1)_k^{d+1}}\left(\sum_{j=0}^{k-1}\frac{1}{m-1+j}-\sum_{j=0}^{k-1}\frac{1}{m+1+j}\right). \tag{3.3}$$

On the other hand, by (3.2) and Lemma 2.4,

$$\Psi_x(0,0)-\Psi_y(0,0)=\frac{(p-1)!}{(1)_{m-2}(1)_m^{d-1}}\left(\frac{1}{m}+\frac{1}{m-1}\right). \tag{3.4}$$

Note that for k among $0,1,\ldots,p-1$, $(1-p)_k\equiv(1)_k \pmod p$ and

$$\frac{(m-1)_k(m+1)_k^{d-1}}{(1)_k^{d+1}}\left(\sum_{j=0}^{k-1}\frac{1}{m-1+j}-\sum_{j=0}^{k-1}\frac{1}{m+1+j}\right)\in\mathbb{Z}_p.$$

This, together with (3.3) and (3.4), gives the desired result. □

Proof of (1.4). For any $x,y\in\mathbb{Z}_p$, let

$$\Phi(x,y)={}_dF_{d-1}\left[\begin{matrix}m-1+x, & m+1+y, & \ldots, & m+1+y\\ & 1, & \ldots, & 1\end{matrix}\,\middle|\,1\right]_{p-1}.$$

Obviously, for any $s,t\in\mathbb{Z}_p$,

$$\Phi(sp,tp)\equiv\Phi(0,0)+sp\Phi_x(0,0)+tp\Phi_y(0,0) \pmod{p^2}. \tag{3.5}$$

In particular,

$$\Phi(-p,0)\equiv\Phi(0,0)-p\Phi_x(0,0) \pmod{p^2}. \tag{3.6}$$

Substituting (3.6) into (3.5), we get

$$\Phi(sp,tp) \equiv \Phi(-p,0) + (s+1)p\Phi_x(0,0) + tp\Phi_y(0,0) \pmod{p^2}.$$

Taking $s = t = -1/d$, in view of Lemmas 2.4 and 3.1,

$$\begin{aligned}
&\Phi\left(-\frac{p}{d}, -\frac{p}{d}\right) \qquad (3.7)\\
&= {}_dF_{d-1}\left[\begin{matrix}\frac{1}{d}-1, 1+\frac{1}{d}, 1+\frac{1}{d}, \ldots, 1+\frac{1}{d}\\ 1, \quad 1, \quad \ldots, \quad 1\end{matrix}\middle| \, 1\right]_{p-1}\\
&\equiv \Phi(-p,0) + \left(1-\frac{1}{d}\right)p\Phi_x(0,0) - \frac{p}{d}\Phi_y(0,0)\\
&\equiv \Phi(-p,0) + \left(1-\frac{1}{d}\right)p\sum_{k=0}^{p-1}\frac{(m-1)_k(m+1)_k^{d-1}}{(1)_k^d}\\
&\quad \left(\sum_{j=0}^{k-1}\frac{1}{m-1+j} - \sum_{j=0}^{k-1}\frac{1}{m+1+j}\right)\\
&\equiv \Phi(-p,0) + \left(1-\frac{1}{d}\right)p\cdot\frac{(p-1)!}{(1)_{m-2}(1)_m^{d-1}}\left(\frac{1}{m}+\frac{1}{m-1}\right) \pmod{p^2}. \qquad (3.8)
\end{aligned}$$

Now we evaluate $\Phi(-p,0)$ modulo p^2. Since $m(d-1) = p+1-m$, by Lemma 2.1,

$$\Phi(-p,0) = (-1)^m\frac{(p+1-m)!}{(1)_m^{d-1}}.$$

With the help of (2.2), we obtain

$$\begin{aligned}
\Phi(-p,0) &= (-1)^m\frac{\Gamma(p-m+2)}{\Gamma(m+1)^{d-1}} = (-1)^{p+2-(m+1)(d-1)}\frac{\Gamma_p(p-m+2)}{\Gamma_p(m+1)^{d-1}}\\
&= (-1)^{1-(m+1)(d-1)}(p-m)(p-m+1)\frac{\Gamma_p(p-m)}{\Gamma_p(m+1)^{d-1}}\\
&= (-1)^m(p-m)(p-m+1)\frac{\Gamma_p(p-m)}{\Gamma_p(m+1)^{d-1}}, \qquad (3.9)
\end{aligned}$$

where in the last step we have used the fact that d is even. In light of (2.3) and Lemma 2.3,

$$\begin{aligned}\frac{\Gamma_p(p-m)}{\Gamma_p(m+1)^{d-1}} &= (-1)^{(m-1)(d-1)}\Gamma_p(p-m)\Gamma_p(-m)^{d-1}\\ &\equiv (-1)^{m-1}\Gamma_p\left(-\frac{1}{d}\right)^d\left(1+\left(1-\frac{1}{d}\right)pG_1\left(-\frac{1}{d}\right)\right)\\ &\quad\left(1-\frac{p}{d}G_1\left(-\frac{1}{d}\right)\right)^{d-1}\\ &\equiv (-1)^{m-1}\Gamma_p\left(-\frac{1}{d}\right)^d \pmod{p^2}. \end{aligned}\tag{3.10}$$

Moreover,

$$\begin{aligned}(p-m)(p-m+1) &= \left(-\frac{1}{d}+\left(1-\frac{1}{d}\right)p\right)\left(1-\frac{1}{d}+\left(1-\frac{1}{d}\right)p\right)\\ &\equiv \frac{1-d}{d^2}+\frac{(d-1)(d-2)p}{d^2} \pmod{p^2}.\end{aligned}$$

Combining this with (3.9) and (3.10), we obtain

$$\Phi(-p,0)\equiv\left(\frac{d-1}{d^2}-\frac{(d-1)(d-2)p}{d^2}\right)\Gamma_p\left(-\frac{1}{d}\right)^d \pmod{p^2}.\tag{3.11}$$

Similarly, it is routine to verify that

$$\begin{aligned}&\left(1-\frac{1}{d}\right)p\cdot\frac{(p-1)!}{(1)_{m-2}(1)_m^{d-1}}\left(\frac{1}{m}+\frac{1}{m-1}\right)\\ &\quad\equiv\frac{(d-1)(d-2)p}{d^2}\Gamma_p\left(-\frac{1}{d}\right)^d \pmod{p^2}.\end{aligned}\tag{3.12}$$

Substituting (3.11) and (3.12) into (3.7), we immediately get (1.4). □

In order to show (1.5), we also need an auxiliary lemma.

Lemma 3.2. *Under the assumptions of Theorem 1.1 (ii), we have*

$$\begin{aligned}&\sum_{k=0}^{p-1}\frac{(m)_k^2(m+1)_k^{d-2}}{(1)_k^d}\left(\sum_{j=0}^{k-1}\frac{1}{m+j}-\sum_{j=0}^{k-1}\frac{1}{m+1+j}\right)\\ &\quad\equiv\frac{(p-1)!}{(1)_{m-1}(1)_m^{d-1}} \pmod{p}.\end{aligned}$$

Proof. For $x,y\in(-1,+\infty)$, set

$$\Upsilon(x,y)={}_{d+1}F_d\left[\begin{matrix}1-p,\ m+x,\ m,\ m+1+y,\ m+1,\ \ldots,\ m+1\\ 1+x,\ \ 1,\ \ 1+y,\ \ 1,\ \ \ldots,\ \ 1\end{matrix}\middle|\ 1\right]_{p-1}.$$

Similarly as in the proof of Lemma 3.1, we are led to the desired result by considering $\Upsilon_x(0,0)-\Upsilon_y(0,0)$. □

Proof of (1.5). For any $x, y \in \mathbb{Z}_p$, set

$$\Omega(x,y) = {}_dF_{d-1}\left[\begin{matrix} m+x, & m+y, & m+1+y, & \ldots, & m+1+y \\ 1, & 1, & \ldots, & & 1 \end{matrix}\middle| 1\right]_{p-1}.$$

Similarly as before, by Lemmas 2.1, 2.4 and 3.2,

$$\begin{aligned} &\Omega\left(-\frac{p}{d}, -\frac{p}{d}\right) \qquad (3.13)\\ &= {}_dF_{d-1}\left[\begin{matrix} \frac{1}{d}, & \frac{1}{d}, & 1+\frac{1}{d}, & 1+\frac{1}{d}, & \ldots, & 1+\frac{1}{d} \\ 1, & 1, & 1, & \ldots, & & 1 \end{matrix}\middle| 1\right]_{p-1}\\ &\equiv \Omega(-p,0) + \left(1-\frac{1}{d}\right) p\Omega_x(0,0) - \frac{p}{d}\Omega_y(0,0)\\ &\equiv \Omega(-p,0) + \left(1-\frac{2}{d}\right) p \sum_{k=0}^{p-1} \frac{(m)_k^2 (m+1)_k^{d-2}}{(1)_k^d} \left(\sum_{j=0}^{k-1}\frac{1}{m+j} - \sum_{j=0}^{k-1}\frac{1}{m+1+j}\right)\\ &\equiv \Omega(-p,0) + \left(1-\frac{2}{d}\right) p \cdot \frac{(p-1)!}{(1)_{m-1}(1)_m^{d-1}} \pmod{p^2}. \qquad (3.14) \end{aligned}$$

Then, in view of Lemmas 2.1–2.3, we have

$$\Omega(-p,0) \equiv \left(-\frac{1}{d^2} + \frac{(d-2)p}{d^2}\right)\Gamma_p\left(-\frac{1}{d}\right)^d \pmod{p^2}$$

and

$$\left(1-\frac{2}{d}\right) p \cdot \frac{(p-1)!}{(1)_{m-1}(1)_m^{d-1}} \equiv \frac{(2-d)p}{d^2}\cdot\Gamma_p\left(-\frac{1}{d}\right)^d \pmod{p^2}.$$

The proof of (1.5) follows by combining the above. □

4. Proof of Theorem 1.2

We need the following identity which can be verified by induction on n.

Lemma 4.1. *For any positive integer n, we have*

$$\sum_{k=0}^{n-1}(4k+1)\frac{\left(\frac{1}{2}\right)_k^2}{k!^2} = \frac{n^2}{4^{2n-1}}\binom{2n}{n}^2.$$

Assuming $p \equiv 1 \pmod 4$ and putting $x = 1/2$ in [6, Corollary 1.4], we have the following result.

Lemma 4.2. *Let $p \equiv 1 \pmod 4$ be a prime and r a positive integer. Then*

$$\sum_{k=0}^{p^r-1}\frac{\left(\frac{1}{2}\right)_k^2}{k!^2} \equiv 1 \pmod{p^2}.$$

Proof of Theorem 1.2. Equivalently, we only need to show

$$\sum_{k=0}^{p^r-1}(4k+1)\frac{\left(\frac{1}{2}\right)_k^2}{k!^2}\equiv p^{2r}\sum_{k=0}^{p^r-1}\frac{\left(\frac{1}{2}\right)_k^2}{k!^2}\pmod{p^{2r+1}}. \tag{4.1}$$

Putting $n=p^{2r}$ in Lemma 4.1, we obtain

$$\sum_{k=0}^{p^r-1}(4k+1)\frac{\left(\frac{1}{2}\right)_k^2}{k!^2}=\frac{p^{2r}}{4^{2p^r-1}}\binom{2p^r}{p^r}^2. \tag{4.2}$$

By Fermat's little theorem,

$$4^{2p^r-1}=4\times 16^{p^r-1}=4\times(16^{p-1})^{\frac{p^r-1}{p-1}}\equiv 4\pmod{p}. \tag{4.3}$$

Using Lucas's theorem, we have

$$\binom{2p^r}{p^r}\equiv\binom{2p^{r-1}}{p^{r-1}}\equiv\cdots\equiv\binom{2}{1}=2\pmod{p}. \tag{4.4}$$

Substituting (4.3) and (4.4) into (4.2), the left-hand side of (4.1) becomes p^{2r} modulo p^{2r+1}. In view of Lemma 4.2, we arrive at Theorem 1.2 at once. □

5. Concluding Remarks

It is routine to check that

$$\begin{aligned}
&{}_dF_{d-1}\left[\begin{matrix}\frac{1}{d}-1, 1+\frac{1}{d}, \ldots, 1+\frac{1}{d}\\ 1, \ldots, 1\end{matrix}\,\middle|\, 1\right]_{p-1}\\
&\quad+(d-1)\,{}_dF_{d-1}\left[\begin{matrix}\frac{1}{d}, \frac{1}{d}, 1+\frac{1}{d}, \ldots, 1+\frac{1}{d}\\ 1, 1, \ldots, 1\end{matrix}\,\middle|\, 1\right]_{p-1}\\
&=d\cdot\,{}_dF_{d-1}\left[\begin{matrix}\frac{1}{d}-1, \frac{1}{d}, 1+\frac{1}{d}, 1+\frac{1}{d}, \ldots, 1+\frac{1}{d}\\ 1, 1, 1, \ldots, 1\end{matrix}\,\middle|\, 1\right]_{p-1}.
\end{aligned} \tag{5.1}$$

Note that Guo [3, Corollaries 4.2 and 4.4] proved that (1.4) and (1.5) also hold for odd integers $d\geq 3$ and even integers $d\geq 4$, respectively. This, together with (1.4), (1.5) and (5.1), gives that

$${}_dF_{d-1}\left[\begin{matrix}\frac{1}{d}-1, \frac{1}{d}, 1+\frac{1}{d}, 1+\frac{1}{d}, \ldots, 1+\frac{1}{d}\\ 1, 1, 1, \ldots, 1\end{matrix}\,\middle|\, 1\right]_{p-1}\equiv 0\pmod{p^2} \tag{5.2}$$

for any integer $d\geq 3$ and prime $p\equiv -1 \pmod{d}$ with $p\neq d-1$. In fact, (5.2) can also be proved independently by using the method we used to prove Theorem 1.1 and the following Karlsson-Minton summation formula:

$${}_{n+1}F_n\left[\begin{matrix}a, b_1+m_1, \ldots, b_n+m_n\\ b_1, \ldots, b_n\end{matrix}\,\middle|\, 1\right]=0$$

provided that $m_1,\ldots,m_n$ are nonnegative integers and $\Re(-a)>m_1+\cdots+m_n$.

Motivated by (5.2) and based on some numerical calculations, we made the following conjecture for further study.

Conjecture 5.1. *Let $d \geq 2$ be an integer. Let n be a positive integer with $n \equiv -1 \pmod{d}$ and $n > d-1$. Then*

$$\frac{(n-1)!^d d^{dn-d}}{n^2} \cdot {}_dF_{d-1}\left[\begin{matrix} \frac{1}{d}-1, & \frac{1}{d}, & 1+\frac{1}{d}, & 1+\frac{1}{d}, & \ldots, & 1+\frac{1}{d} \\ & 1, & 1, & 1, & \ldots, & 1 \end{matrix}\middle| \, 1\right]_{n-1} \in \mathbb{Z}. \tag{5.3}$$

We think it is possible that the 'creative microscoping' could be used to prove Conjecture 5.1. We hope that an interested reader will make some progress on it.

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